SPRINT 3

CarClinch Dealership Assistant

GROUP 4 | CST8922 | 03/30/2026

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Project Overview

Goal

Build an AI-powered email assistant that helps car dealerships automatically respond to leads and schedule appointments using GPT-4.1 mini.

Sprint 3

- ➔ Sentiment/intent analysis layer
- ➔ Escalation logic
- ➔ Followup emails
- ➔ Appointment booking
- ➔ Infrastructure security (Terraform)

ON TRACK

Member Updates: Alice & Aryan

Alice

Followup Emails, Sentiment Analysis, Prompt Tuning

Delivered

- Designed **followup** email trigger conditions w/ **Durable Timer**
- GPT-4.1 mini **prompt finetuning**
- **Sentiment/intent analysis** layer before response generation

Working On

- **Unit tests** for followup emails, escalation, & analysis layer into prod
- **Edge case** handling for escalation & followup

Blocker

- Waiting for team to review PR before moving forward

Next Steps

- Complete analysis layer & unit tests
- Communicate with Desmond on prompt finetuning completion and handover code refactoring

Aryan

Escalation Logic, Terraform Security

Delivered

- Implemented **escalation logic** for high-urgency leads to dealerships
- Reinforced **Terraform** config for **security** gaps in Azure resource deployment

Working On

- Final project plan research & documentation

Blocker

- N/A

Next Steps

- Lock down Terraform security configs for staging/prod environment
- Final project plan research & documentation

Member Updates: Desmond & Junfeng

Desmond

Prompt Finetuning, Code Refactoring, QA

Delivered

- ➔ Completed QA testing for all Sprint 2 features
- ➔ Identified code readability issues for handover
- ➔ Identified prompt finetuning requirements for behavior

Working On

- ➔ Refactoring codebase for efficiency & maintainability for client handover
- ➔ Final project plan & documentation

Blocker

- ➔ N/A

Next Steps

- ➔ Finalize refactoring form services for client review and handover
- ➔ Validate prompt outputs against edge-case scenarios
- ➔ Final project plan & documentation

Junfeng

Appointment Booking, AI Agent

Delivered

- ➔ Designed appointment booking feature architecture
- ➔ Identified AI agent architecture to strength booking capabilities

Working On

- ➔ Implementing core appointment booking functionality
- ➔ Implementing AI agent workflow into existing LLM functionality

Blocker

- ➔ Calendar integration API spec pending finalization of feasibility
- ➔ Determine if AI agent is too large for existing scope

Next Steps

- ➔ Deliver appointment booking MVP
- ➔ Integrate AI agent workflow with email-processing-service

Member Updates: Saizal

Saizal

Documentation, Final Project Plan

Delivered

- ➔ Updated technical documentation for Sprint 2 features
- ➔ Contributed to sprint review and retrospective notes

Working On

- ➔ Starting final documentation for full project
- ➔ Drafting final project plan structure and content outline

Blocker

- ➔ Needs feature freeze from team to finalize docs

Next Steps

- ➔ Final project plan research
- ➔ Document deployment procedures and system architecture
- ➔ Coordinate with all members for distributing project plan responsibilities

Issues & Roadblocks

Issue/Blocker	Status	Action / Resolution
Edge cases for followup email and escalation emails	Open	➔ Add unit tests to validate ➔ Ensure that all messages (lead intake and replies) are forwarded ➔ Ensure followups are not spammed outside of reasonable context
“Insufficient quota” error blocked Azure OpenAI deployment	Resolved	➔ Switched to Microsoft Foundry for GPT-4.1 mini
AI agent refactoring within scope	In Progress	➔ Junfeng implementing booking MVP alongside AI agent ➔ Will move forward without AI agent if not possible before Wednesday

Risks & Dependencies

Incomplete Edge Cases for Followup & Escalation

Comprehensive test coverage & peer review before merge into production to ensure that no edge cases are missed.

Impact: High | Prob: Medium

Poor Sentiment/Intent Accuracy

Prompt with varied test scenarios and tune escalation thresholds conservatively to ensure that the AI cannot respond to any risky scenario.

Impact: Medium | Prob: Medium

Azure OpenAI Quota Issues

Resolved by migrating to Microsoft Foundry. Monitoring quota usage proactively from here on out to see if quota issues show up again.

Impact: High | Prob: Low

Timeline & Milestones

Wk 5	✓ Design core email system
Wk 6	✓ Form frontend/backend + email processing service
Wk 7–8	✓ Refactor email handling (ACS → SMTP/IMAP)
Wk 9	✓ GPT-4.1 mini integration + Terraform to Azure
Wk 10–11	▶ **Sentiment/intent, escalation logic, followup emails**
Wk 12	○ Appointment booking + AI agent workflow
Wk 13	○ Contingency buffer
Wk 14	○ Finalize deliverable, presentation & handover